LECTURE L17-18 · MODULE 3 · CO3

Missing data

Week 9 | Slides: `DATA209_Module_3_Data_Quality_Engineering.pptx`, pages 12–21 | Laboratory: **P17-18**

TOPICS COVERED

- MCAR, MAR, MNAR
- Detection methods including disguised missingness
- Imputation: mean/median, KNN, regression-based

Missingness is data

The pattern of what is absent often carries more information than the values would have. Before imputing anything, ask **why** the value is missing — the answer determines whether any imputation is legitimate.

Three mechanisms

- **MCAR — missing completely at random.** Absence is unrelated to anything. A tube was dropped.

Deletion is unbiased but wasteful; imputation is safe.

- **MAR — missing at random.** Absence depends on *observed* variables. Older patients skipped a

test. Imputation conditioned on those variables works.

- **MNAR — missing not at random.** Absence depends on the *unobserved value itself*. High earners

hide income. No imputation is unbiased; the missingness must be modelled.

You cannot prove MCAR, nor distinguish MAR from MNAR, from the data alone. You argue for a mechanism from domain knowledge.

Evidence you can gather

Create a binary indicator for missingness in each column and explore it like any other variable. Correlate it with the observed columns — a strong relationship argues against MCAR. Compare the distribution of other variables between the missing and non-missing groups. Visualise the missingness matrix: block patterns reveal structural causes such as a form change. Little's test gives formal evidence against MCAR, but not a MAR/MNAR distinction.

Best of all: **ask the data owner why the field is blank.** This beats every statistical test.

Keep the indicator. It costs one column, preserves the information you are about to overwrite, and documents your assumption.

Disguised missingness

`pima.csv` reports zero nulls. Yet 374 records (48.7%) have Insulin recorded as 0, and 227 (29.6%) have SkinThickness as 0 — and a living person has neither. Sentinel values defeat every automatic null check. This is the single most important idea in the module.

The menu, and its costs

STRATEGY	USE WHEN	COST
Listwise deletion	MCAR, few rows affected	Pima loses 49% of rows to Insulin alone
Column deletion	extreme missingness, weak column	loses a possibly predictive feature
Mean / median	quick baseline, low missingness	shrinks variance, distorts correlations
KNN	MAR, features genuinely related	needs scaling; slow on large data
Iterative / MICE	MAR, several columns affected	complex; can leak if fitted on all data
Indicator + fill	missingness itself is informative	one extra column each

What imputation costs

Mean imputation gives every missing row an identical value, so those rows contribute **zero deviation** from the mean and the standard deviation must fall. In the controlled experiment in P17-18 — where values are deleted with the truth retained — mean imputation lands 1,094 units from the truth on average and shrinks the standard deviation by 6.5%; KNN lands 517 units away and shrinks it by 1.2%. The dataset looks more precise than it is.

Rules

Fit the imputer on the **training data only**, after the split, inside a pipeline. Compare distributions before and after and report the change in mean, standard deviation and skew. Try at least two strategies. State the assumed mechanism in writing: *"median imputation, assuming MAR given Age and BMI"* is a complete claim.

KEY TERMS

`MCAR` `MAR` `MNAR` `Disguised missingness` `Sentinel value` `Listwise deletion` `Missingness indicator` `KNN imputation` `MICE` `Little's test`

COMMON ERRORS TO AVOID

- Trusting isna() to find all missing values
- Imputing before the train/test split
- Reporting an imputation without measuring the change in variance
- Choosing a strategy without stating the assumed mechanism

READINGS — ALL FREE TO ACCESS

1. pandas — working with missing data
   `https://pandas.pydata.org/docs/user_guide/missing_data.html`
2. scikit-learn — imputation of missing values
   `https://scikit-learn.org/stable/modules/impute.html`
3. Kuhn & Johnson — Feature Engineering and Selection
   `http://www.feat.engineering/`

TEST YOURSELF

1. Income is missing more often for high earners. Which mechanism, and what follows?
2. Why does mean imputation shrink the standard deviation?
3. Name two pieces of evidence that argue against MCAR.
4. Why must the imputer be fitted after the split rather than before?